

Week 18: Authentication & Authorization

1. JWT Implementation

JWT (JSON Web Token) is used for secure authentication in web applications. It generates a token after successful login, which is stored and verified for protected routes. JWT improves session handling, API security, and stateless authentication.

2. Register/Login API

Register and login APIs are essential for user authentication. Registration stores new user details in the database, while login verifies user credentials before granting access. These APIs provide secure user management and authentication flow.

3. Bcrypt for Password Hashing

Bcrypt is used to hash passwords before storing them in the database. It converts plain-text passwords into encrypted form, making them more secure. Even if the database is exposed, hashed passwords remain protected.

4. Role-Based Access Control (RBAC)

Role-based access control is used to manage permissions based on user roles like admin, user, or moderator. It ensures only authorized users can access specific routes, dashboards, or actions. This improves security and system control.

5. Protected Routes & Middleware

Protected routes prevent unauthorized access to sensitive pages or APIs. Middleware is used to verify JWT tokens before processing requests. This strengthens application security.

Conclusion

This week focused on authentication and authorization. By learning JWT, login/register APIs, bcrypt hashing, RBAC, and protected routes, strong application security and user management skills were developed.